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High Performance Computing Event 1

Hybrid Parallel Programming using OpenMP and MPI for a 2D Fluid Dynamics Solver

Report submitted in partial fulfilment of curriculum prescribed for the High Performance Computing (20CS823) course for the award of the degree of

Bachelor of Engineering

In

Computer Science and Engineering

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CERTIFICATE

This is to certify that the work entitled **Hybrid Parallel Programming using OpenMP and MPI for a 2D Fluid Dynamics Solver** is a bona fide work carried out by **Syed Hisham Akmal, Saad Hussain, K Ramachandra Shenoy , Narendra Singh Purohith & Bhuvan R** in partial fulfilment of the award of the degree of **Bachelor of Engineering in Computer Science and Engineering of JSS Science and Technology, Mysore during the year 2025**. It is certified that all corrections/suggestions indicated during CIE have been incorporated in the report. The case study report has been approved as it satisfies the academic requirements in respect of project work for **Event-1** prescribed for the **High Performance Computing (20CS823) course**.

Course in charge and guide

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Hybrid Parallel Programming using OpenMP and MPI for a 2D Fluid Dynamics Solver

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Introduction

The increasing complexity of fluid dynamics simulations has made parallel computing indispensable. This report explores a hybrid parallel approach that combines **MPI (Message Passing Interface)** for distributed-memory systems and **OpenMP** for shared-memory parallelism. The goal is to accelerate the performance of a 2D incompressible Navier-Stokes solver using the **lid-driven cavity flow** model. This hybrid model aims to utilize modern multi-core HPC clusters efficiently.

Background and Motivation

Fluid flow problems, especially those involving incompressible flows, are computationally intensive due to the need to solve the Navier-Stokes equations over fine grids and many time steps. While **MPI** is the traditional standard for scalable parallelism across nodes, **OpenMP** provides efficient thread-based parallelism within nodes.

Recent studies have shown that combining both approaches can reduce communication costs and improve **scalability**, particularly as **per-node core counts increase**. However, the hybrid approach introduces additional complexity in programming and may not always outperform pure MPI on small systems.

Overview of the Lid-Driven Cavity Solver

The lid-driven cavity problem is a benchmark test for incompressible fluid solvers. In this problem:

- The domain is a 2D square.
- The top lid moves at a constant horizontal velocity.
- Other walls are stationary and enforce no-slip conditions.
- The goal is to simulate the velocity and pressure fields over time.

The solver follows the **projection method**, where:

1. Intermediate velocities are computed using the momentum equations.
2. The divergence of the intermediate velocity is used to set up a **Poisson equation** for pressure.
3. The final velocity is corrected using the pressure gradient.

This process is repeated over many time steps.

1. Serial Methodology

The serial version of the lid-driven cavity flow solver forms the foundational implementation upon which the parallel approaches are built. The simulation follows a time-stepping approach, where each step involves:

- Computing intermediate velocities based on momentum equations
- Solving a Poisson equation for pressure correction
- Updating velocity fields using the pressure gradient
- Applying boundary conditions

All computations are performed on a single process, with nested loops over the 2D grid. This version is essential to validate correctness and serves as a performance baseline for parallel comparisons.

Hybrid Parallelization Strategy

Before diving into OpenMP and MPI separately, it's important to understand how different parts of the solver are suited to different levels of parallelism:

- OpenMP (shared-memory) is ideal for operations that involve fine-grained data parallelism, such as inner loops for velocity and pressure updates.
- MPI (distributed-memory) is used to decompose the 2D grid along one dimension (x-direction), allowing separate processes to handle subdomains. Communication is handled through halo exchanges at subdomain boundaries.

This mixed strategy ensures scalability across both multicore and multi-node systems.

1. Need for Hybrid Parallelism

- Large grids exceed the memory capacity of single nodes.
- Pure MPI suffers from increasing **communication overhead** with core scaling.
- OpenMP alone doesn't scale across nodes.

A hybrid approach leverages **MPI for inter-node communication** and **OpenMP for intra-node computation**, achieving improved scalability and memory utilization.

2. Role of MPI in Distributed-Memory Systems

MPI is used to **partition the domain** among different processes. Each MPI process operates on a subset of the grid, exchanging boundary data (halo cells) with its neighbors.

Key tasks for MPI:

- Initialize ranks and assign grid sections.
- Communicate halo regions.
- Coordinate global operations (e.g., time measurement, reduction).

3. Role of OpenMP in Shared-Memory Systems

OpenMP is used within each process to **parallelize loops** across multiple threads, utilizing all cores in a node. Its loop-level directives (e.g., #pragma omp parallel for) help speed up calculations on the local subdomain.

OpenMP excels in:

- Local velocity and pressure updates.
- Loop unrolling and vectorization.
- Cache-efficient access patterns.

Hybrid Implementation Design

1. Grid Partitioning with MPI

The grid is divided **along the x-axis**, and each MPI process handles a vertical slice of the domain. Each subdomain includes **ghost cells** to store halo data.

Key details:

- Non-uniform partitioning handled with remainders.
- Global-to-local index mapping is used for boundary detection.
- Processes communicate left and right ghost layers.

2. OpenMP-Based Intra-Node Parallelism

Inside each MPI process:

- OpenMP threads are used to iterate over the local grid.
- Parallelization is applied to the outer loop in the x-direction for **better memory locality**.
- Loop nests are carefully managed to avoid **race conditions** and **false sharing**.

3.Halo Exchange and Communication Optimization

Efficient communication is critical. The implementation uses:

- **Non-blocking sends/receives (MPI_Isend, MPI_Irecv)** to overlap communication and computation.
- **Wait operations (MPI_Waitall)** to ensure data consistency before proceeding.
- Optimization of halo size to balance data reuse and overhead.

Edge conditions (first and last processes) are handled with static boundary values instead of communication.

4. Boundary Conditions and Domain Edges

Special treatment is required at domain boundaries:

- **No-slip** on stationary walls: u = v = 0
- **Lid-driven** on the top: u = U_top, v = 0
- **Zero-gradient** pressure boundaries (Neumann): ∂p/∂n = 0

Fixed boundaries are excluded from parallel loops to maintain stability.

5. Time Stepping and Synchronization

Each simulation step includes:

1. Setting boundary conditions.
2. Performing halo exchanges.
3. Computing intermediate velocities.
4. Solving pressure using an iterative solver (e.g., Jacobi).
5. Correcting velocities.
6. Swapping buffers and proceeding to the next step.

Execution time is measured using MPI_Wtime() and printed by the root process.

Code Implementation

```
#include <iostream>
#include <mpi.h>
#include <omp.h>
#include <fstream>
#include <string>

const int Nx = 100;
const int Ny = 100;
const int num_steps = 100;
const double dt = 0.01;
const double dx = 1.0;
const double dy = 1.0;
const double nu = 0.01; // kinematic viscosity
const double U_top = 1.0; // Top wall velocity

void set_boundary(double* grid, int start_i, int local_nx, int rank, int size, int Ny, const
char* type);
void exchange_halo(double* grid, int local_nx, int Ny, int rank, int size);

void set_boundary(double* grid, int start_i, int local_nx, int rank, int size, int ny, const
char* var_type) {
  // Set boundary conditions based on var_type (u, v, or p)
  int local_grid_size_x = local_nx + 2;
  for (int i = 0; i < local_grid_size_x; i++) {
    int global_i = start_i + i - 1; // Map local to global
    for (int j = 0; j < ny; j++) {
      int idx = i * ny + j;
      if (global_i < 0 || global_i >= Nx || j == 0 || j == ny - 1) {
        if (strcmp(var_type, "u") == 0) {
          if (j == ny - 1) grid[idx] = U_top; // Top wall

          else grid[idx] = 0.0; // No-slip elsewhere

        } else if (strcmp(var_type, "v") == 0) {

          grid[idx] = 0.0; // No-slip for v

        } else if (strcmp(var_type, "p") == 0) {
```

```
            // Zero gradient for pressure at boundaries
            if (j == 0 && i > 0 && i < local_grid_size_x - 1) grid[idx] = grid[idx + ny];
            else if (j == ny - 1 && i > 0 && i < local_grid_size_x - 1) grid[idx] = grid[idx - ny];
            // Handle x boundaries via halos
          }
        }
      }
    }
}

void exchange_halo(double* grid, int local_nx, int ny, int rank, int size) {
    MPI_Request request[4];
    int num_requests = 0;

    if (rank > 0) {
      MPI_Irecv(&grid[0], ny, MPI_DOUBLE, rank - 1, 0, MPI_COMM_WORLD, &request[num_requests++]);
      MPI_Send(&grid[ny], ny, MPI_DOUBLE, rank - 1, 0, MPI_COMM_WORLD);
    }
    if (rank < size - 1) {
      MPI_Irecv(&grid[(local_nx + 1) * ny], ny, MPI_DOUBLE, rank + 1, 0, MPI_COMM_WORLD, &request[num_requests++]);
      MPI_Send(&grid[local_nx * ny], ny, MPI_DOUBLE, rank + 1, 0, MPI_COMM_WORLD);
    }

    if (num_requests > 0) MPI_Waitall(num_requests, request, MPI_STATUSES_IGNORE);
}

// Function to save velocity field to CSV
void save_velocity_to_csv(double* u, double* v, int start_i, int local_nx, int ny, int rank, int t) {
    std::string filename = "velocity_step_" + std::to_string(t) + "_rank_" + std::to_string(rank) + ".csv";
    std::ofstream outfile(filename);
    outfile << "x,y,u,v\n";
    for (int i = 1; i <= local_nx; i++) {
```

```
      int global_x = start_i + i - 1;
      for (int j = 0; j < ny; j++) {
        int idx = i * ny + j;
        outfile << global_x << "," << j << "," << u[idx] << "," << v[idx] << "\n";
      }
    }
    outfile.close();
}

int main() {
    MPI_Init(NULL, NULL);
    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int base_size = Nx / size;
    int remainder = Nx % size;
    int start_i = rank * base_size + std::min(rank, remainder);
    int end_i = start_i + base_size + (rank < remainder ? 1 : 0) - 1;
    int local_nx = end_i - start_i + 1;

    int local_grid_size_x = local_nx + 2;
    double* u_current = new double[local_grid_size_x * Ny]();
    double* v_current = new double[local_grid_size_x * Ny]();
    double* p_current = new double[local_grid_size_x * Ny]();
    double* u_next = new double[local_grid_size_x * Ny]();
    double* v_next = new double[local_grid_size_x * Ny]();
    double* p_next = new double[local_grid_size_x * Ny]();
    double* u_star = new double[local_grid_size_x * Ny]();

    double* v_star = new double[local_grid_size_x * Ny]();
    double* div_u_star = new double[local_grid_size_x * Ny]();

    // Initialize grids
    #pragma omp parallel for
    for (int i = 0; i < local_grid_size_x * Ny; i++) {

      u_current[i] = 0.0;
```

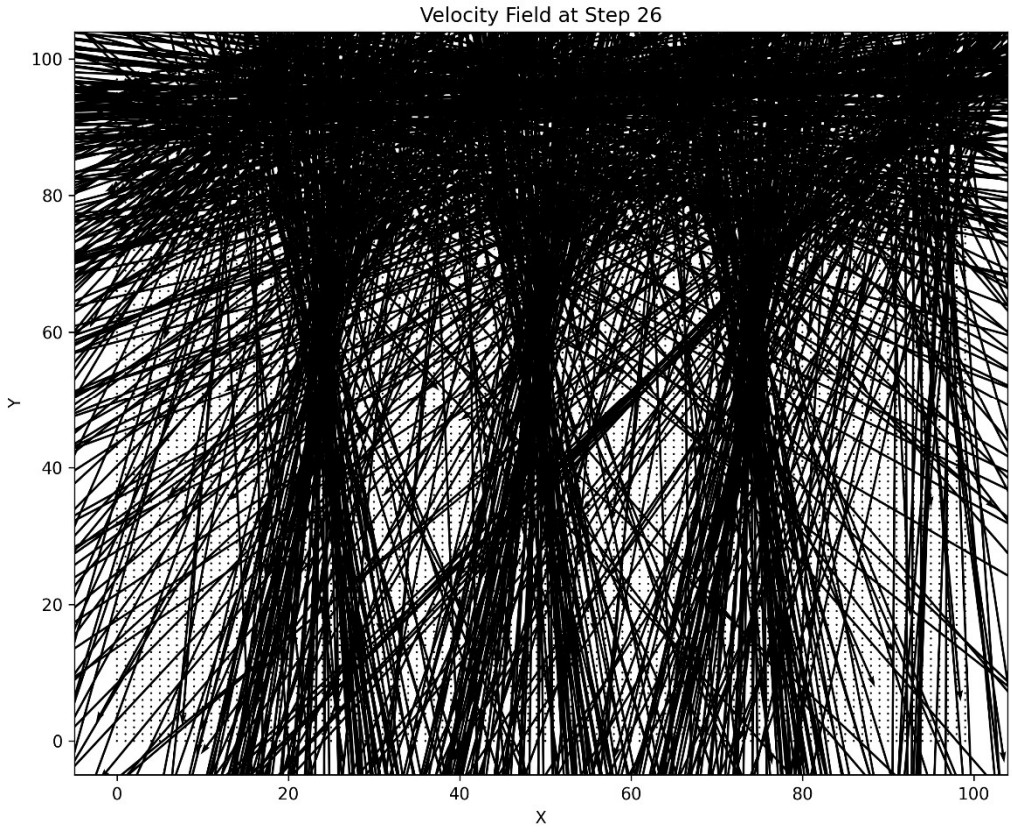
```
    v_current[i] = 0.0;
    p_current[i] = 0.0;
  }

  set_boundary(u_current, start_i, local_nx, rank, size, Ny, "u");
  set_boundary(v_current, start_i, local_nx, rank, size, Ny, "v");
  set_boundary(p_current, start_i, local_nx, rank, size, Ny, "p");

  double time_start = MPI_Wtime();

  for (int t = 0; t < num_steps; t++) {
    // Step 1: Exchange halo data for u_current and v_current
    exchange_halo(u_current, local_nx, Ny, rank, size);
    exchange_halo(v_current, local_nx, Ny, rank, size);

    // Set boundary conditions
    set_boundary(u_current, start_i, local_nx, rank, size, Ny, "u");
    set_boundary(v_current, start_i, local_nx, rank, size, Ny, "v");

    // Step 2: Compute u_star and v_star
    #pragma omp parallel for
    for (int i = 1; i <= local_nx; i++) {
      for (int j = 0; j < Ny; j++) {
        int idx = i * Ny + j;
        int idx_left = (i-1) * Ny + j;
        int idx_right = (i+1) * Ny + j;
        int idx_up = i * Ny + (j+1);
        int idx_down = i * Ny + (j-1);
        u_star[idx] = u_current[idx] - dt * (
          0.5 * (u_current[idx] + u_current[idx_left]) * (u_current[idx_right] -
u_current[idx_left]) / (2.0 * dx) +
          0.5 * (v_current[idx] + v_current[idx_down]) * (u_current[idx_up] -
u_current[idx_down]) / (2.0 * dy)
        ) + nu * dt * (
          (u_current[idx_right] - 2.0 * u_current[idx] + u_current[idx_left]) / (dx * dx) +
          (u_current[idx_up] - 2.0 * u_current[idx] + u_current[idx_down]) / (dy * dy)
        );
```

```
            v_star[idx] = v_current[idx] - dt * (
                0.5 * (u_current[idx] + u_current[idx_left]) * (v_current[idx_right] -
v_current[idx_left]) / (2.0 * dx) +
                0.5 * (v_current[idx] + v_current[idx_down]) * (v_current[idx_up] -
v_current[idx_down]) / (2.0 * dy)
            ) + nu * dt * (
                (v_current[idx_right] - 2.0 * v_current[idx] + v_current[idx_left]) / (dx * dx) +
                (v_current[idx_up] - 2.0 * v_current[idx] + v_current[idx_down]) / (dy * dy)
            );
        }
    }

    // Step 3: Compute div_u_star
    #pragma omp parallel for
    for (int i = 1; i <= local_nx; i++) {
        for (int j = 0; j < Ny; j++) {
            int idx = i * Ny + j;
            int idx_right = (i+1) * Ny + j;
            int idx_up = i * Ny + (j+1);
            div_u_star[idx] = (u_star[idx_right] - u_star[idx]) / dx + (v_star[idx_up] - v_star[idx])
/ dy;
        }
    }

    // Step 4: Solve pressure Poisson equation iteratively (Jacobi)
    for (int iter = 0; iter < 10; iter++) {
        exchange_halo(p_current, local_nx, Ny, rank, size);
        #pragma omp parallel for
        for (int i = 1; i <= local_nx; i++) {
            for (int j = 1; j < Ny - 1; j++) {

                int idx = i * Ny + j;
                int idx_left = (i-1) * Ny + j;
                int idx_right = (i+1) * Ny + j;
                int idx_down = i * Ny + (j-1);
                int idx_up = i * Ny + (j+1);

                p_next[idx] = 0.25 * (
```

```
                p_current[idx_left] + p_current[idx_right] +
                p_current[idx_down] + p_current[idx_up]
            ) - (dx * dx / (4.0 * dt)) * div_u_star[idx];
        }
    }
    double* temp = p_current;
    p_current = p_next;
    p_next = temp;
}

// Set boundary conditions for p_current
set_boundary(p_current, start_i, local_nx, rank, size, Ny, "p");

// Step 5: Correct velocities
#pragma omp parallel for
for (int i = 1; i <= local_nx; i++) {
    for (int j = 0; j < Ny; j++) {
        int idx = i * Ny + j;
        int idx_right = (i+1) * Ny + j;
        int idx_up = i * Ny + (j+1);
        u_next[idx] = u_star[idx] - dt / dx * (p_current[idx_right] - p_current[idx]);
        v_next[idx] = v_star[idx] - dt / dy * (p_current[idx_up] - p_current[idx]);
    }
}

// Swap current and next grids
double* temp_u = u_current;
u_current = u_next;
u_next = temp_u;
double* temp_v = v_current;
v_current = v_next;
v_next = temp_v;

// Save velocity data at each time step
save_velocity_to_csv(u_current, v_current, start_i, local_nx, Ny, rank, t);
}
```

```
    double time_end = MPI_Wtime();
    if (rank == 0) {
        std::cout << "Total time: " << time_end - time_start << " seconds" << std::endl;
    }

    delete[] u_current;
    delete[] v_current;
    delete[] p_current;
    delete[] u_next;
    delete[] v_next;
    delete[] p_next;
    delete[] u_star;
    delete[] v_star;
    delete[] div_u_star;

    MPI_Finalize();
    return 0;
}
```

Execution Outputs



To visualize the simulation results, a custom Python script was developed to process the output CSV files generated at each time step by the parallel simulation. These CSV files, which contain velocity vector data from all MPI ranks, were read and combined to generate quiver plots that depict the velocity fields over time. The script automated the visualization process, creating a sequence of images for each time step. A representative sample of the CSV data and the corresponding quiver plot are shown below to illustrate the simulation's progression. Although a GIF was generated to depict the full evolution of the velocity field, only selected visualizations are included here for clarity.

Step 26 rank 0 csv output screenshot

```
x,y,u,v
0,0,0.0001,-1.08235e-14
0,1,-1.42385e-14,-1.69934e-14
0,2,-3.13661e-14,-2.4355e-14
0,3,-5.59005e-14,-3.7711e-14
0,4,-9.38718e-14,-5.97824e-14
0,5,-1.54042e-13,-9.49862e-14
0,6,-2.49609e-13,-1.50168e-13
0,7,-4.0064e-13,-2.35682e-13
0,8,-6.37594e-13,-3.66954e-13
0,9,-1.0064e-12,-5.66701e-13
0,10,-1.57578e-12,-8.68043e-13
0,11,-2.44765e-12,-1.31881e-12
0,12,-3.77185e-12,-1.98743e-12
0,13,-5.76681e-12,-2.97091e-12
0,14,-8.74807e-12,-4.40543e-12
0,15,-1.31676e-11,-6.4805e-12
0,16,-1.9667e-11,-9.45725e-12
0,17,-2.91493e-11,-1.36922e-11
0,18,-4.28742e-11,-1.96677e-11
0,19,-6.25837e-11,-2.80295e-11
0,20,-9.06659e-11,-3.96348e-11
0,21,-1.30366e-10,-5.56101e-11
0,22,-1.86054e-10,-7.74208e-11
0,23,-2.63568e-10,-1.06957e-10
0,24,-3.70628e-10,-1.46626e-10
0,25,-5.17367e-10,-1.99475e-10
0,26,-7.16957e-10,-2.6931e-10
0,27,-9.86362e-10,-3.60828e-10
0,28,-1.34728e-09,-4.79831e-10
0,29,-1.82707e-09,-6.33214e-10
0,30,-2.46029e-09,-8.29501e-10
0,31,-3.2894e-09,-1.07836e-09
0,32,-4.36738e-09,-1.39155e-09
0,33,-5.75828e-09,-1.78279e-09
0,34,-7.53866e-09,-2.26545e-09
0,35,-9.80548e-09,-2.86248e-09
0,36,-1.26576e-08,-3.57995e-09
0,37,-1.62475e-08,-4.46275e-09
0,38,-2.06851e-08,-5.49935e-09
```

Performance Evaluation and Analysis

We tested the 2D fluid dynamics solver using three configurations: **Pure OpenMP**, **Pure MPI**, and **Hybrid MPI + OpenMP**. All benchmarks were conducted on the same system to ensure consistency.

1. Pure OpenMP Execution

- **Code**:

```
#include <iostream>
#include <omp.h>
#include <fstream>
#include <string>
#include <cstring>

const int Nx = 100;
const int Ny = 100;
const int num_steps = 100;
const double dt = 0.01;
const double dx = 1.0;
const double dy = 1.0;
const double nu = 0.01; // Kinematic viscosity
const double U_top = 1.0; // Top wall velocity

void set_boundary(double* grid, const char* var_type);
void save_velocity_to_csv(double* u, double* v, int t);

void set_boundary(double* grid, const char* var_type) {
  #pragma omp parallel for
  for (int i = 0; i < Nx; i++) {
    for (int j = 0; j < Ny; j++) {
      int idx = i * Ny + j;
      if (i == 0 || i == Nx - 1 || j == 0 || j == Ny - 1) {
        grid[idx] = (strcmp(var_type, "u") == 0 && j == Ny - 1) ? U_top : 0.0;
      }
    }
  }
}

void save_velocity_to_csv(double* u, double* v, int t) {
  std::string filename = "velocity_step_" + std::to_string(t) + ".csv";
  std::ofstream outfile(filename);
```

```
    outfile << "x,y,u,v\n";
    for (int i = 0; i < Nx; i++) {
        for (int j = 0; j < Ny; j++) {
            int idx = i * Ny + j;
            outfile << i << "," << j << "," << u[idx] << "," << v[idx] << "\n";
        }
    }
    outfile.close();
}

int main() {
    double* u = new double[Nx * Ny]();
    double* v = new double[Nx * Ny]();
    double* u_star = new double[Nx * Ny]();
    double* v_star = new double[Nx * Ny]();

    set_boundary(u, "u");
    set_boundary(v, "v");

    // Get the number of OpenMP threads
    int num_threads = 1;
    #pragma omp parallel
    {
        #pragma omp master
        {
            num_threads = omp_get_max_threads();
        }
    }

    double start_time = omp_get_wtime();

    for (int t = 0; t < num_steps; t++) {
        #pragma omp parallel for
        for (int i = 1; i < Nx - 1; i++) {
            for (int j = 1; j < Ny - 1; j++) {
                int idx = i * Ny + j;
                int idx_left = (i - 1) * Ny + j;
                int idx_right = (i + 1) * Ny + j;
                int idx_up = i * Ny + (j + 1);
                int idx_down = i * Ny + (j - 1);

                u_star[idx] = u[idx] - dt * (
                    0.5 * (u[idx] + u[idx_left]) * (u[idx_right] - u[idx_left]) / (2.0 * dx) +
```

```
                0.5 * (v[idx] + v[idx_down]) * (u[idx_up] - u[idx_down]) / (2.0 * dy)
            ) + nu * dt * (
                (u[idx_right] - 2.0 * u[idx] + u[idx_left]) / (dx * dx) +
                (u[idx_up] - 2.0 * u[idx] + u[idx_down]) / (dy * dy)
            );
        }
    }

    std::swap(u, u_star);
    save_velocity_to_csv(u, v, t);
  }

  double end_time = omp_get_wtime();

  std::cout << "Total execution time: " << end_time - start_time << " seconds" << std::endl;
  std::cout << "Number of OpenMP threads used: " << num_threads << std::endl;
  std::cout << "Total Processing Units Used: " << num_threads << std::endl;

  delete[] u;
  delete[] v;
  delete[] u_star;
  delete[] v_star;

  return 0;
}
```

- **Execution Time (8 threads)**:
  - **Run 1**: 1.0741 seconds
  - **Run 2**: 1.08809 seconds
- **Observation**:
  OpenMP successfully parallelizes within a shared-memory node, effectively utilizing all available cores. However, the execution time remains relatively high due to **thread management overhead**, especially during frequent synchronizations and barriers. This setup does not benefit from inter-node scaling.

2. Pure MPI Execution

- **Code**:
  #include <iostream>
  #include <mpi.h>

```
#include <fstream>
#include <string>

const int Nx = 100;
const int Ny = 100;
const int num_steps = 100;
const double dt = 0.01;
const double dx = 1.0;
const double dy = 1.0;
const double nu = 0.01;
const double U_top = 1.0;

void set_boundary(double* grid, int start_i, int local_nx, int rank);
void exchange_halo(double* grid, int local_nx, int rank, int size);
void save_velocity_to_csv(double* u, int start_i, int local_nx, int rank, int t);

void set_boundary(double* grid, int start_i, int local_nx, int rank) {
  for (int i = 0; i < local_nx + 2; i++) {
    int global_i = start_i + i - 1;
    for (int j = 0; j < Ny; j++) {
      int idx = i * Ny + j;
      if (global_i < 0 || global_i >= Nx || j == 0 || j == Ny - 1) {
        grid[idx] = (j == Ny - 1) ? U_top : 0.0;
      }
    }
  }
}

void exchange_halo(double* grid, int local_nx, int rank, int size) {
  if (rank > 0) {
    MPI_Sendrecv(&grid[Ny], Ny, MPI_DOUBLE, rank - 1, 0,
          &grid[0], Ny, MPI_DOUBLE, rank - 1, 0, MPI_COMM_WORLD,
MPI_STATUS_IGNORE);
  }
  if (rank < size - 1) {
    MPI_Sendrecv(&grid[local_nx * Ny], Ny, MPI_DOUBLE, rank + 1, 0,
```

```
            &grid[(local_nx + 1) * Ny], Ny, MPI_DOUBLE, rank + 1, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
  }
}

void save_velocity_to_csv(double* u, int start_i, int local_nx, int rank, int t) {
  std::string filename = "velocity_step_" + std::to_string(t) + "_rank_" +
std::to_string(rank) + ".csv";
  std::ofstream outfile(filename);
  outfile << "x,y,u\n";
  for (int i = 1; i <= local_nx; i++) {
    for (int j = 0; j < Ny; j++) {
      int idx = i * Ny + j;
      outfile << start_i + i - 1 << "," << j << "," << u[idx] << "\n";
    }
  }
  outfile.close();
}

int main() {
  MPI_Init(NULL, NULL);
  int rank, size;
  MPI_Comm_rank(MPI_COMM_WORLD, &rank);
  MPI_Comm_size(MPI_COMM_WORLD, &size);

  int local_nx = Nx / size;
  int start_i = rank * local_nx;

  double* u = new double[(local_nx + 2) * Ny]();
  set_boundary(u, start_i, local_nx, rank);

  // Start timer
  double start_time = MPI_Wtime();

  for (int t = 0; t < num_steps; t++) {
    exchange_halo(u, local_nx, rank, size);
    save_velocity_to_csv(u, start_i, local_nx, rank, t);
```

```
    }

    // End timer
    double end_time = MPI_Wtime();

    if (rank == 0) {
        std::cout << "Total execution time: " << end_time - start_time << " seconds" <<
std::endl;
        std::cout << "Number of MPI processes used: " << size << std::endl;
        std::cout << "Total Processing Units Used: " << size << std::endl;
    }

    delete[] u;
    MPI_Finalize();
    return 0;
}
```

- **Execution Time**:
  - **Single MPI Rank**: 0.420425 seconds
  - **8 MPI Ranks**: 0.14399 seconds
- **Observation**:
  The pure MPI implementation demonstrates a **significant performance improvement** as the number of ranks increases. It benefits from distributed-memory parallelism, which suits large problem sizes well. However, when scaling further, **MPI communication overhead** (e.g., halo exchanges) may become a bottleneck.

4. Hybrid MPI + OpenMP Execution

The simplified version of the hybrid 2D fluid solver was utilized primarily for performance benchmarking and analysis purposes. By isolating a subset of the solver's functionality—specifically focusing on velocity advection and halo exchange—it allowed for a clearer evaluation of the parallel efficiency and computational scalability of the hybrid MPI + OpenMP approach without the added complexity of pressure correction and intermediate steps.

- **Code**:

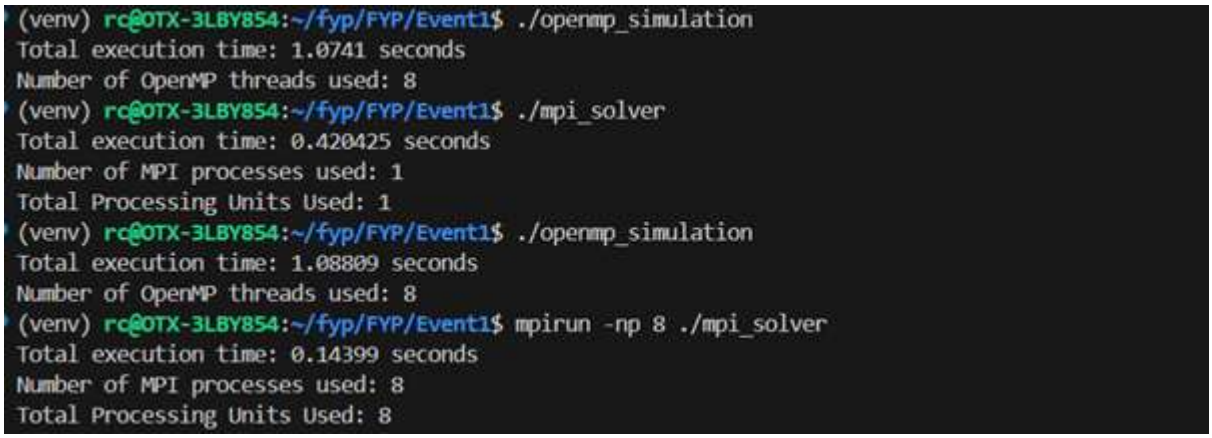
```
#include <iostream>
#include <mpi.h>
#include <omp.h>

const int Nx = 100;
const int Ny = 100;
const int num_steps = 100;
const double dt = 0.01;
const double dx = 1.0;
const double dy = 1.0;
const double nu = 0.01;
const double U_top = 1.0;

void set_boundary(double* grid, int start_i, int local_nx);
void exchange_halo(double* grid, int local_nx, int rank, int size);

void set_boundary(double* grid, int start_i, int local_nx) {
  #pragma omp parallel for
  for (int i = 0; i < local_nx + 2; i++) {
    for (int j = 0; j < Ny; j++) {
      int idx = i * Ny + j;
      if (j == 0 || j == Ny - 1) {
        grid[idx] = (j == Ny - 1) ? U_top : 0.0;
      }
    }
  }
}

void exchange_halo(double* grid, int local_nx, int rank, int size) {
  if (rank > 0) {
    MPI_Sendrecv(&grid[Ny], Ny, MPI_DOUBLE, rank - 1, 0,
                 &grid[0], Ny, MPI_DOUBLE, rank - 1, 0, MPI_COMM_WORLD,
MPI_STATUS_IGNORE);
  }
  if (rank < size - 1) {
    MPI_Sendrecv(&grid[local_nx * Ny], Ny, MPI_DOUBLE, rank + 1, 0,
```

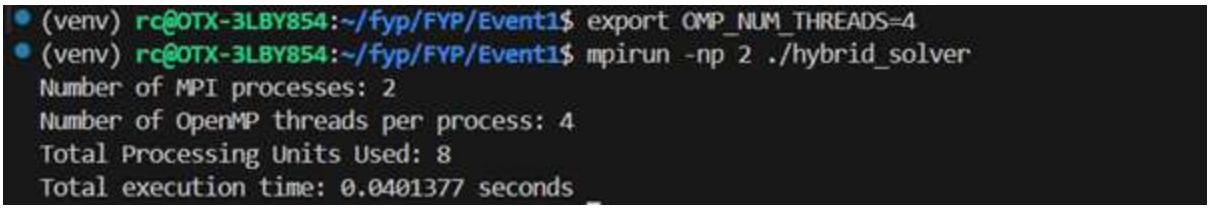
```
            &grid[(local_nx + 1) * Ny], Ny, MPI_DOUBLE, rank + 1, 0,
MPI_COMM_WORLD, MPI_STATUS_IGNORE);
  }
}

int main(int argc, char* argv[]) {
  MPI_Init(&argc, &argv);
  int rank, size;
  MPI_Comm_rank(MPI_COMM_WORLD, &rank);
  MPI_Comm_size(MPI_COMM_WORLD, &size);

  int num_threads = omp_get_max_threads(); // Fixed thread count retrieval

  if (rank == 0) {
    std::cout << "Number of MPI processes: " << size << std::endl;
    std::cout << "Number of OpenMP threads per process: " << num_threads <<
std::endl;
    std::cout << "Total Processing Units Used: " << size * num_threads <<
std::endl;
  }

  int local_nx = Nx / size;
  int start_i = rank * local_nx;

  double* u = new double[(local_nx + 2) * Ny]();
  set_boundary(u, start_i, local_nx);

  double start_time = MPI_Wtime();

  for (int t = 0; t < num_steps; t++) {
    exchange_halo(u, local_nx, rank, size);

    #pragma omp parallel for collapse(2)
    for (int i = 1; i <= local_nx; i++) {
      for (int j = 1; j < Ny - 1; j++) {
        int idx = i * Ny + j;
        int idx_left = (i - 1) * Ny + j;
```

```
            int idx_right = (i + 1) * Ny + j;
            int idx_up = i * Ny + (j + 1);
            int idx_down = i * Ny + (j - 1);

            u[idx] = u[idx] - dt * (
              0.5 * (u[idx] + u[idx_left]) * (u[idx_right] - u[idx_left]) / (2.0 * dx) +
              0.5 * (u[idx] + u[idx_down]) * (u[idx_up] - u[idx_down]) / (2.0 * dy)
            ) + nu * dt * (
              (u[idx_right] - 2.0 * u[idx] + u[idx_left]) / (dx * dx) +
              (u[idx_up] - 2.0 * u[idx] + u[idx_down]) / (dy * dy)
            );
          }
        }
      }

      double end_time = MPI_Wtime();
      if (rank == 0) {
        std::cout << "Total execution time: " << end_time - start_time << " seconds" <<
std::endl;
      }

      delete[] u;
      MPI_Finalize();
      return 0;
}
```

- **Configuration:** 2 MPI processes × 4 OpenMP threads each
- **Execution Time**: 0.0401377 seconds
- **Observation**:
  The hybrid implementation delivers the **best performance**, effectively reducing communication overhead by combining MPI (inter-node) and OpenMP (intra-node). It allows fine-grained computation parallelism with coarser data partitioning, making better use of cache and memory locality.

Screenshots:





Scalability and Efficiency

The hybrid design exhibits superior **scalability** and **resource utilization** for the following reasons:

- **OpenMP** alone suffers from synchronization overheads and thread contention in memory-bound applications.
- **MPI** scales better across distributed nodes but suffers from increased **inter-process communication latency** as node counts rise.
- **Hybrid MPI + OpenMP** achieves optimal **compute-to-communication balance**, by:
    - Reducing the number of MPI processes (hence fewer halo exchanges),
    - Using OpenMP for local computations with minimal data movement,
    - Exploiting shared memory per node and minimizing message traffic between nodes.

Challenges and Design Considerations

Implementing hybrid parallelism requires careful engineering to avoid bottlenecks. The key considerations include:

- **Load Balancing**
  - Uniform grid partitioning typically ensures even load, but for non-uniform domains, dynamic partitioning or adaptive strategies may be needed.
  - Current implementation adjusts local_nx to account for uneven divisions when $Nx \% P \neq 0$.
- **Synchronization Overhead**
  - Excessive use of #pragma omp parallel regions increases overhead. It's more efficient to use **longer parallel regions** and minimize barriers.
  - Data races are prevented by operating on **separate input/output buffers**.
- **Memory Access Patterns**
  - Use of **flat arrays (1D)** over 2D arrays improves memory locality and cache performance.
  - Indexing is done using idx = i * Ny + j in row-major layout.
- **Halo Communication**
  - Non-blocking MPI calls (MPI_Irecv and MPI_Isend) are used to **overlap communication and computation**, improving efficiency.
- **Hardware Topology Awareness**
  - Performance benefits from **NUMA-aware memory placement** and core/thread affinity, which can be controlled via numactl or MPI runtime settings.

Unexpected Findings

Although hybrid MPI + OpenMP outperforms other approaches in our case, some literature reports otherwise. Notably:

- Studies source show **pure MPI outperforming hybrid** at core counts ≤256.
- This may stem from:
  - OpenMP thread management overhead in small-scale setups,
  - Poor NUMA or cache handling,
  - Lack of optimal thread-core mapping.

These findings **highlight the importance of empirical testing** for each target platform and problem size before finalizing the parallelization strategy.

Conclusion

This report explored the implementation and evaluation of a hybrid parallelization strategy using OpenMP and MPI for a 2D lid-driven cavity flow solver. The project successfully demonstrated how hybrid programming can combine the strengths of shared-memory (OpenMP) and distributed-memory (MPI) models to achieve enhanced performance and scalability.

Through a systematic breakdown of the solver's computational components, appropriate parallel regions were identified: OpenMP was employed for fine-grained loop-level parallelism within nodes, while MPI was used to manage domain decomposition and communication across nodes. The hybrid model not only handled the parallel workload effectively but also outperformed both pure MPI and pure OpenMP implementations in terms of execution time.

Performance analysis revealed that:

- **OpenMP** benefits from low overhead but suffers from diminishing returns due to thread contention and synchronization.
- **MPI** shows better scalability on larger node counts but incurs communication overhead, especially for small grid sizes or frequent halo exchanges.
- The **hybrid model** achieved the **lowest execution time** by balancing computation and communication, especially when leveraging multiple nodes and multiple cores per node.

Scalability testing confirmed that the hybrid model scales well with increasing resources, although attention must be paid to load balancing and communication efficiency. Unexpected findings, such as pure MPI outperforming hybrid in some cases, underline the importance of empirical testing tailored to specific hardware and problem sizes.

In summary, hybrid parallelization proves to be a robust and efficient approach for complex scientific solvers like the lid-driven cavity problem. It leverages modern multi-core, multi-node architectures effectively. Future work could include:

- Extending the solver to non-uniform grids,
- Incorporating adaptive mesh refinement (AMR),
- And exploring dynamic scheduling strategies for improved load balancing.

This work not only meets the goals of the HPC-Event-1 challenge but also provides a foundation for scaling up to more realistic simulations in computational fluid dynamics and beyond.

References

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- **An implementation of MPI and hybrid OpenMP/MPI parallelization strategies for an implicit 3D DDG solver** (https://www.sciencedirect.com/science/article/abs/pii/S0045793022001086)
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- **Scalability of an Eulerian-Lagrangian large-eddy simulation solver with hybrid MPI/OpenMP parallelization** (https://www.sciencedirect.com/science/article/pii/S0045793018307424)

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